

Osmoregulated Periplasmic Glucan Biosynthesis (UPA00637): OpgH and OpgG

A review-style synthesis anchored on the Pseudomonas putida KT2440 OpgH/OpgG module, set in its broader proteobacterial context.

1. Executive summary

Osmoregulated periplasmic glucans (OPGs; historically "membrane-derived oligosaccharides," MDOs, in *Escherichia coli*) are a family of β -glucose polymers that accumulate to high concentration in the periplasm of Gram-negative bacteria when the surrounding medium is dilute (low osmolarity). The UniPathway module **UPA00637** captures the conserved, minimal core of their biosynthesis: an inner-membrane glucosyltransferase, **OpgH**, that polymerizes a β -glucan backbone from UDP-glucose on the cytoplasmic face of the inner membrane, and a periplasmic partner protein, **OpgG**, that is strictly required for producing mature backbone. In *P. putida* KT2440 these are encoded by *opgH* (PSEPK OpgH) and *opgG* (PSEPK OpgG), the orthologues of the *E. coli* *mdoGH/opgGH* bicistronic operon.

The system is best understood as a **two-protein glucan factory** that is gated by osmolarity and fed by central carbon metabolism. Its output is not a fixed molecule but a tunable one: the backbone is β -1,2-linked with β -1,6 branches (in *E. coli* and pseudomonads), and it is decorated post-synthetically by accessory enzymes (succinyl, phosphoglycerol, O-acetyl groups) that are **not** part of UPA00637 proper. OpgH additionally "moonlights" as a nutrient-dependent inhibitor of cell division by sequestering FtsZ — a role mechanistically separate from glucan synthesis and unique to OpgH within this module. Loss of OPGs produces a strikingly pleiotropic envelope phenotype (motility loss, exopolysaccharide overproduction, biofilm and virulence defects), much of it transduced through the Rcs envelope-stress phosphorelay.

The strongest, best-replicated claims are: (i) *opgG* and *opgH* are individually essential for backbone synthesis; (ii) the donor is UDP-glucose and the process is osmoregulated; and (iii) OpgH sequesters FtsZ. The most important open questions concern the precise biochemical

division of labor between OpgH and OpgG, whether *P. putida* makes linear or cyclic glucans, and how much of the phenotypic pleiotropy is a direct glucan effect versus an indirect signaling consequence.

2. Definition and biological boundaries

2.1 What is included

UPA00637 as scoped here comprises exactly two obligatory steps:

1. **Inner-membrane OPG glucosyltransferase step — OpgH.** A large polytopic cytoplasmic-membrane protein (~97 kDa in *E. coli* MdoH) carrying glucosyltransferase activity; it transfers glucosyl units from UDP-glucose to build the β -glucan backbone.
2. **Periplasmic OPG biosynthesis step — OpgG.** A ~56 kDa periplasmic protein (carbohydrate-binding role) that is obligatorily required for backbone production, presumably acting on the nascent/translocated glucan on the periplasmic side.

Together they generate the **unsubstituted glucose backbone** of OPGs. This is the appropriate boundary of the "biosynthesis of OPG backbone" pathway.

2.2 What is adjacent and should be treated separately

- **Backbone modification enzymes (OpgB, OpgC, OpgD).** These add or edit substituents or influence chain length but are not backbone-synthesizing: *opgB* (sn-1-phosphoglycerol transferase), *opgC* (succinyl/O-acyl transferase), and *opgD* (an OpgG paralog affecting chain length). In *Shigella flexneri* the *opg* family is *opgGH* (bicistronic) plus *opgB*, *opgC*, and *opgD* [PMID 20062978]; in *Rhodobacter sphaeroides* the operon is *opgGIH* with a downstream *opgC* controlling succinylation [PMID 12027885]. These modifications determine anionic charge and are physiologically important but mechanistically downstream of UPA00637.
- **Alternative, non-homologous OPG routes (Ndv/Chv and Cgs systems).** The Rhizobiaceae synthesize **cyclic β -(1,2)-glucans** via a single very large (~319 kDa) inner-membrane protein (NdvB/ChvB) that forms a protein-linked oligosaccharide intermediate [PMID 8830704]. In *Brucella*, the equivalent cyclic- β -1,2-glucan is made by a dedicated inner-membrane **synthase/transporter/modifier complex (Cgs/Cgt/Cgm)** that physically assembles (homo- and heterotypic interactions, polar localization) to couple synthesis, periplasmic transport, and succinylation [PMID 25733613]. Both achieve a similar biological

outcome (periplasmic osmoprotectant/virulence glucan) by a completely different molecular means and are **not** homologous to OpgGH. They should not be conflated with the Opg pathway.

- **"Orphan" cyclic- β -glucan synthases in pseudomonads.** A recently described BcsA-like GT2/GH17 fusion protein present in *P. aeruginosa*, *P. fluorescens*, *P. putida* KT2440, and *P. syringae* is predicted to make **cyclic β -(1,3)-glucan**, a distinct polysaccharide from OpgGH-dependent OPGs [PMID 37267309]. Its existence means "periplasmic glucan" in *P. putida* is not synonymous with the OpgGH product; care is needed to avoid attributing cyclic- β -glucan chemistry to the OpgGH module.
- **NdvB-mediated biofilm antibiotic resistance.** In *P. aeruginosa*, *ndvB* (PA1163) contributes to biofilm antibiotic tolerance but is explicitly **not** responsible for the OpgGH-dependent OPG described at the *opgGH* locus [PMID 17906125]. A frequent source of confusion.
- **Downstream regulatory consequences.** Colanic-acid capsule synthesis, Rcs phosphorelay activation, and motility are affected *by* OPG status but are separate systems [PMID 9352941].

2.3 Competing definitions

"OPG," "MDO," "periplasmic glucan," and "cyclic glucan" are used loosely in the older literature. Bohin's classification organizes OPGs into **four families based on backbone linkage/branching** [PMID 10779706]; the OpgGH-dependent family (β -1,2 backbone with β -1,6 branches) is Family I / the enterobacterial-pseudomonad type. Terminological drift — especially calling all periplasmic glucans "MDO" or lumping Ndv/Chv cyclic glucans with OpgGH glucans — remains a real hazard.

3. Mechanistic overview

3.1 Best current model of the sequence of events

1. **Substrate supply (accessory/upstream).** UDP-glucose is generated by central metabolism (e.g., *galU/galF*). It is the glucosyl donor for OPG synthesis [PMID 26150539].
2. **Backbone polymerization at the inner membrane (obligatory — OpgH).** OpgH, a polytopic inner-membrane glucosyltransferase, uses UDP-glucose to build the β -glucan backbone. Genetic and biochemical evidence in *E. coli* shows OpgH/MdoH is required for glucosyltransferase activity [PMID 7934824], and the reaction uses **acyl carrier protein (ACP) as a cofactor** [PMID 26150539]. Polymerization is thought to occur on the

cytoplasmic face with concomitant/subsequent translocation of the growing glucan to the periplasm.

3. **Periplasmic maturation (obligatory — OpgG).** OpgG, a periplasmic β -sandwich/carbohydrate-binding protein, is strictly required for producing the mature backbone; loss of OpgG abolishes glucan even when OpgH is intact [PMID 11325942]. Its precise catalytic/chaperone role (chain-length control, branching, or scaffolding of OpgH) is not fully resolved.
4. **Substitution/editing (conditional/accessory).** OpgB (phosphoglycerol), OpgC (succinyl/O-acetyl), and OpgD (chain length) act on the periplasmic backbone, tuning charge and size [PMID 20062978;

P 12027885

These are species-dependent and dispensable for making glucose backbone.

3.2 Obligatory vs conditional vs accessory

- **Obligatory:** OpgH (backbone glucosyltransferase) and OpgG (periplasmic maturation). Either knockout abolishes OPGs [PMID 11325942;

P 19118363

P 20062978

- **Conditional:** osmoregulatory input (transcriptional repression at high osmolarity plus enzyme-level control) [PMID 17906125;

P 10779706

— synthesis is strongly favored at low osmolarity but the genes are present regardless.

- **Accessory:** substitution enzymes (OpgB/C), chain-length modulator (OpgD), UDP-glucose supply, and the **OpgH–FtsZ moonlighting activity** (affects cell size, not glucan chemistry) [PMID 23935518].

3.3 Regulatory logic

Osmoregulation is dual-level: high osmolarity represses *opgGH* transcription and also dampens enzyme activity; low osmolarity derepresses and activates, matching glucan output to periplasmic osmotic demand, with feedback control preventing overaccumulation [PMID 10779706]. Periplasmic OPG levels are, in turn, read out by the **Rcs (RcsC/RcsB/RcsA) phosphorelay**, which controls colanic-acid capsule and other envelope genes [PMID 9352941].

4. Major molecular players and active assemblies

Player	Location	Role	Status in UPA00637	Key evidence
OpgH (MdoH; HrpM in <i>P. syringae</i>)	Inner membrane, polytopic (~97 kDa)	Glucosyltransferase; builds β -glucan backbone from UDP-glucose; moonlights as FtsZ-sequestering cell-size regulator	Core, obligatory	P 7934824 ; 26150539; 23935518
OpgG (MdoG)	Periplasm (~56 kDa)	Carbohydrate-binding; required for mature backbone (chain length/branching/scaffolding)	Core, obligatory	P 7934824 ; 11325942
OpgI	(in some α -proteobacteria, e.g. <i>R. sphaeroides</i> <i>opgGIH</i>)	Additional backbone-synthesis factor	Accessory/ lineage-specific	P 12027885
OpgD	Periplasm, OpgG paralog	Chain-length control	Accessory	P 20062978
OpgB	Membrane	sn-1-phosphoglycerol transferase (substitution)	Accessory	P 20062978
OpgC	Membrane	Succinyl/O-acyl transferase (anionic substitution)	Accessory	P 12027885 ; 20062978
UDP-glucose (GalU/ GalF)	Cytoplasm	Glucosyl donor	Substrate (upstream)	P 26150539
ACP	Cytoplasm	Cofactor for the glucosyltransferase reaction	Cofactor	P 26150539

Active assembly. The functional unit is an OpgH–OpgG couple spanning the inner membrane and periplasm: OpgH polymerizes and (likely) translocates, OpgG completes maturation in the periplasm. There is no strong evidence that they form an obligate stable heterocomplex vs. a functional/kinetic coupling; this remains an open structural question. Note that OpgH's cell-division role involves a *distinct* interaction with **FtsZ** at the septum [PMID 23935518], a second "assembly" entirely separate from the glucan machinery.

5. Evolutionary and cell-biological variation

5.1 Across lineages

- **OPGs (or functional equivalents) are near-universal in Proteobacteria.** OPGs occur in almost all α -, β - and γ -Proteobacteria, and where classical OPGs are absent (ϵ -Proteobacteria) OPG-like oligosaccharides take their place [PMID 26265506] — underscoring a pervasive, conserved requirement for a periplasmic glucan in the Gram-negative envelope.
- **OpgGH is the deepest, most widely distributed OPG-synthesis solution in Proteobacteria** — the *opgG/opgH* pair is the most broadly conserved OPG-biosynthetic gene set in proteobacterial genomes [PMID 17906125]. The best representatives for understanding the ancestral OpgGH role are the well-characterized enterobacterial (*E. coli* MdoGH) and pseudomonad (*P. aeruginosa* OpgGH; *P. syringae* HrpM) systems, which retain the minimal bicistronic architecture. *P. putida* KT2440 OpgH/OpgG belong to this conserved core.
- **Structural backbone families vary** [PMID 10779706]: enterobacteria/pseudomonads make β -1,2 backbones with β -1,6 branches; *P. aeruginosa* OPGs are **linear** β -1,2 glucans with a few β -1,6 branches and succinyl substituents [PMID 17906125]; *R. sphaeroides* makes **cyclic anionic** OPGs [PMID 12027885]. Thus even within the OpgGH family, linear vs cyclic and neutral vs anionic outcomes differ by lineage.
- **Convergent, non-homologous alternatives exist:** Rhizobiaceae use NdvB/ChvB (~319 kDa, protein-linked intermediate) to make cyclic β -1,2-glucans [PMID 8830704]; *Brucella* uses a Cgs/Cgt/Cgm inner-membrane complex for the same class of molecule [PMID 25733613]; and some pseudomonads additionally carry an orphan GT2/GH17 synthase predicted to make cyclic β -1,3-glucan [PMID 37267309]. These are alternative routes to a periplasmic osmoprotectant glucan.

5.2 Physiological-state and cell-cycle variation

- **Osmotic state** is the dominant variable: OPGs accumulate at low osmolarity and are downregulated at high osmolarity [PMID 17906125;

P 10779706

- **Nutrient/growth-rate state** is coupled to OpgH's cell-size function: under nutrient-rich, fast growth, OpgH (sensing UDP-glucose) localizes to the septum and delays division, yielding larger cells [PMID 23935518].
- **Lineage-specific loss.** *Yersinia pestis* lost the *opgGH* operon during its emergence from *Y. pseudotuberculosis*; the loss correlates with reduced cell size, and restoring *opgGH* restores the larger size — a natural experiment showing the operon is dispensable in an obligate mammal/flea lifestyle [PMID 26150539]. Intriguingly, *Y. pseudotuberculosis* retains the operon but does **not** detectably make OPGs under tested conditions, showing that gene presence $\neq$ glucan production.

5.3 A note on the convergent cell-size module

OpgH is functionally analogous to *Bacillus subtilis* **UgtP**, a glucosyltransferase that also inhibits FtsZ in a growth-rate-dependent manner — yet OpgH and UgtP share **no homology and different enzymatic activities**, a textbook case of convergent evolution using UDP-glucose as a shared nutrient proxy [PMID 23935518]. This cell-size role is a **later elaboration** layered onto the ancestral glucan-synthesis function of OpgH and is absent from OpgG.

6. Constraints, dependencies, and failure modes

- **Compartment/order constraints.** Backbone polymerization must precede periplasmic maturation and any substitution; substitutions (OpgB/C) act on periplasmic glucan and therefore cannot occur before OpgH-dependent synthesis and translocation. UDP-glucose must be available in the cytoplasm before OpgH can act [PMID 26150539].
- **Non-redundancy.** OpgG and OpgH cannot substitute for one another: both are required, ruling out models in which OpgH alone suffices to make mature glucan [PMID 11325942;

P 20062978

- **Osmotic gating.** High osmolarity is a physiological "off" state; OPG synthesis at high osmolarity is minimized by combined transcriptional and enzymatic repression [PMID 10779706].
- **Failure modes (opg-null phenotype).** Loss of OPGs causes a highly pleiotropic envelope disturbance: EPS overproduction, reduced motility, bile-salt hypersensitivity, reduced secreted enzymes, and virulence loss in phytopathogens [PMID 11325942]; low-osmolarity growth/motility defects and ~100-fold virulence attenuation in *Salmonella* [PMID 19118363]; altered biofilm architecture in *P. aeruginosa* [PMID 17906125]. Much of this is transduced by Rcs, which senses periplasmic glucan status [PMID 9352941]. Importantly, the *specific* role differs by species: OPGs act as periplasmic signaling molecules that indirectly regulate virulence genes, but when secreted they can directly sequester antibiotics or engage host cells during invasion [PMID 26265506] — so "the OPG phenotype" is context-dependent rather than a single universal function.
- **Evidence ruling out alternatives.** The *opgGH* locus, not *ndvB*, is responsible for OpgGH-type OPGs in *P. aeruginosa* [PMID 17906125]; and OpgH's FtsZ role is separable from glucan output (Yersinia data), so the two OpgH functions must be considered independently [PMID 26150539];

P 23935518

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7. Controversies and open questions

1. **Division of labor between OpgH and OpgG.** OpgH clearly carries glucosyltransferase activity, but OpgG's exact biochemical role — chain-length determination, branching, scaffolding/chaperoning of OpgH, or a hydrolase/transglycosylase activity — remains unresolved. It is "required," but the mechanism of that requirement is inferred, not proven [PMID 7934824];

P 11325942

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2. **Linear vs cyclic product in *P. putida*.** *P. aeruginosa* OpgGH makes linear glucans [PMID 17906125], but pseudomonads also harbor an orphan synthase predicted to make cyclic β -1,3-glucan [PMID 37267309]. The exact structure of the *P. putida* KT2440 OpgGH product and how it partitions from any orphan-synthase product has not, to our knowledge, been directly determined — a genuine gap for the scoped organism.

3. **How direct is the pleiotropy?** Because Rcs reads periplasmic glucan status [PMID 9352941], many *opg*-null phenotypes may be indirect signaling consequences rather than direct structural effects of missing glucan. Disentangling direct vs signaling effects is unresolved.
4. **Gene presence vs glucan production.** *Y. pseudotuberculosis* keeps *opgGH* but makes no detectable OPGs under tested conditions [PMID 26150539], implying additional, uncharacterized activation requirements (substrate flux, cofactor, or unknown regulator). This warns against inferring glucan production from genome annotation alone — directly relevant to a genome-annotation-driven module like *P. putida* UPA00637.
5. **Cross-organism extrapolation.** Much mechanistic detail comes from *E. coli*, with structural/product detail from *P. aeruginosa*, *R. sphaeroides*, and enteric pathogens. Extending these to *P. putida* is reasonable given *OpgGH* conservation but should be flagged as inference, not measurement, for KT2440 specifically.

8. Key references

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Uncertainty statement

Claims about OpgH backbone glucosyltransferase activity, UDP-glucose donor usage, osmoregulation, the opg-null pleiotropic phenotype, and OpgH-FtsZ sequestration are **strongly supported** by multiple organisms and direct experiments. The precise catalytic role of **OpgG**, the exact **structure of the *P. putida* KT2440 OpgGH product** (linear vs cyclic; substitution pattern), and the extent to which the *P. putida* module behaves as its *E. coli/P. aeruginosa* orthologues do are **inferences** from homology and neighboring-organism data, and should be treated as provisional for KT2440 specifically.

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